

AMENDMENT NO. 3 MAY 2006
TO
IS 101 (PART 5/SEC 2) : 1988 METHODS OF
SAMPLING AND TEST FOR PAINTS, VARNISHES AND
RELATED PRODUCTS

PART 5 MECHANICAL TESTS
Section 2 Flexibility and Adhesion
(Third Revision)

(Page 1, clause 1) — Substitute the following for the existing clause:

‘1 Scope — Prescribes the following five tests for determination of flexibility and adhesion:

- a) Bend test,
- b) Scratch hardness test,
- c) Cupping test,
- d) Pull off test, and
- e) Cross cut adhesion test.’

(Page 12, clause 5.6.2, read with Amendment No. 1) — Add the following at the end:

‘6 CROSS CUT ADHESION TEST

6.1 Outline of the Method — This method is applicable for the assessment of the adhesion of the coating films to metallic substrates by applying and removing the 25 mm semi-transparent pressure sensitive tape over cuts made in the film.

6.2 Test Methods — There are essentially two test methods for the determination of adhesion.

6.2.1 Test Method 1 — It is generally used for job at site and if the dry film thickness is 125 microns or more.

6.2.2 Test Method 2 — It is generally used for job in the laboratory and if the dry film thickness is less than 125 microns.

6.2.3 These methods do not distinguish between higher levels of adhesion. These methods are not applicable in multicoated system where failure may occur between coats.

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6.3 Test Method 1

6.3.1 Apparatus

6.3.1.1 Cutting tool — Sharp razor blade, scalpel, knife or other cutting edges are in good condition.

6.3.1.2 Cutting guide steel or other hard metal straightedge to ensure cuts.

6.3.1.3 Tape — 25 mm wide semi-transparent pressure sensitive tape (*see* IS 13262 : 1992 Pressure sensitive adhesive tapes with plastic base).

6.3.1.4 Eraser — attached at the end of the pencil.

6.3.1.5 Illumination — A light source to determine whether the cuts have been made through the film to the substrate.

6.3.1.6 Test specimen — In the field the specimen is the coated structure or article on which the adhesion is to be evaluated. In the laboratory, apply the materials to mild steel panels or panel of the composition and surface conditions on which it is desired to determine adhesion.

6.3.2 Procedure — Select a clean and dry area of film free from blemishes and minor surface imperfection. Two cuts are made in the film about 40-mm long that intersect near their middle with a smaller angle of 30° to 45°. When making the incisions, use the straightedge and cut through the coating to the substrate in one steady motion. Inspect the incisions for reflection of light from the metal substrate to establish that the coating film has been penetrated. If the substrate has not been reached make another cuts in different location. Do not attempt to deepen the previous cut as this may affect adhesion along the incision. Remove two complete laps of the pressure-sensitive tape from the roll. An additional length of 75 mm is removed at a steady rate. Place the centre of the tape at the intersection of the cuts with the tape running in the same direction as the smaller angles. Smooth the tape into place by finger in the area of the incisions and then rub firmly with the eraser on the end of the pencil. The colour under the transparent tape is useful indication of when good contact has been made. Within 90 ± 30 seconds of application, remove the tape by seizing the free end by pulling it off rapidly back upon itself as close to an angle of 180° as possible. Inspect the cut area for removal of coating from the substrate or previous coating. The coated panel is considered to satisfy the test if no peeling or removal of the film is observed.

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6.4 Test Method 2

6.4.1 Apparatus

6.4.1.1 Cutting tool — Sharp razor blade, scalpel, knife or other cutting device having a cutting edge angle between 15 and 30 degrees that will make either a single cut or several cuts at once.

6.4.1.2 Cutting guide — Steel or other hard metal straightedge to ensure cuts.

6.4.1.3 Tape — 25 mm wide semi-transparent pressure sensitive tape (see IS 13262 : 1992).

6.4.1.4 Eraser — attached at the end of the pencil.

6.4.1.5 Illumination — A light source to determine whether the cuts have been made through the film to the substrate.

6.4.1.6 Magnifying glass

6.4.1.7 Test specimen — In the field the specimen is the coated structure or article on which the adhesion is to be evaluated. In the laboratory, apply the materials to mild steel panels or panel of the composition and surface conditions on which it is desired to determine adhesion.

6.4.2 Procedure — Select a clean and dry area of film free from blemishes and minor surface imperfection. Place the panel on a firm base and under the illuminated magnifier make parallel cuts. For coatings having a drying film thickness up to and including 50 micron space the cuts 1 mm apart and make eleven cuts. For coatings having a drying film thickness between 50 micron and 125 micron space the cuts 2 mm apart and make six cuts. For film thickness greater than 125 microns this method is not applicable. Make all cuts about 20 mm long. Cut through the film to the substrate in one steady motion using just sufficient pressure on the cutting tool to have the cutting edge reach the substrate. After making the required cuts brush the film slightly with a soft brush to remove any detached flakes or ribbon of coatings. An additional number of cuts at 90° centered on the original cuts are made. Brush the area as before and inspect the incision for reflection of light from the substrate. If the metal has not been reached make another grid in a different location. Remove two complete laps of the pressure-sensitive tape from the roll. An additional length of 75 mm is removed at a steady rate. Place the centre of the tape over grid. Smooth the tape into place by finger in the area of the incisions and then

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rub firmly with the eraser on the end of the pencil. The colour under the transparent tape is useful indication of when good contact has been made.

Within 90 ± 30 seconds of application, remove the tape by seizing the free end by pulling it off rapidly back upon itself as close to an angle of 180° as possible. Inspect the grid area for removal of the coating from the substrate or from a previous coating using the illuminated magnifier. The edges of the cuts should be smooth and none of the square of the lattice should be detached. The coated panel is considered to satisfy the test if no peeling or removal of the film is observed.

(CHD 20)